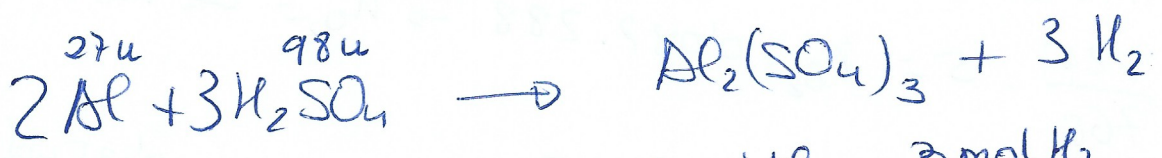


EJERCICIO 11

ESTEQUIOMETRÍA



$$X \text{ g} \quad n_{\text{Zn}} = \frac{X}{65'4} \quad n_{\text{H}_2} = n_{\text{Zn}} = \frac{X}{65'4} \text{ mol}$$



$$(0'156 - X) \text{ g} \\ n_{\text{Al}} = \frac{0'156 - X}{27}$$

$$\frac{2 \text{ mol Al}}{(0'156 - X) / 27} = \frac{3 \text{ mol H}_2}{n_{\text{H}_2}}$$

$$n_{\text{H}_2} = \left(\frac{0'156 - X}{27} \right) \cdot \frac{3}{2} = \frac{0'156 - X}{18} \text{ mol}$$

El H₂ obtenido proviene de la reacción del Zn más el de la reacción de Al

$$P.V = nRT \quad \frac{725}{760} \cdot 0'114 = n_{\text{H}_2} \cdot 0'082 \cdot 300$$

$$n_{\text{H}_2} = 0'0044 \text{ mol totls}$$

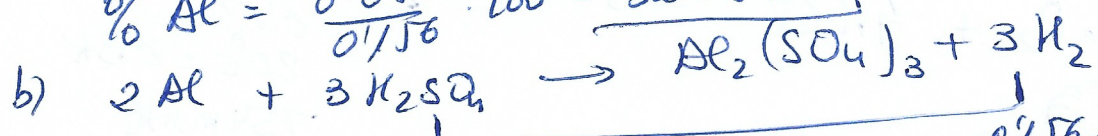
$$\frac{X}{65'4} + \frac{0'156 - X}{18} = 0'004 \rightarrow 18X + 10'2 - 65'4X = 5'18$$

$$5'02 = 47'4X \rightarrow X = 0'106 \text{ g de Zn}$$

$$0'156 - X = 0'156 - 0'106 = 0'05 \text{ g de Al}$$

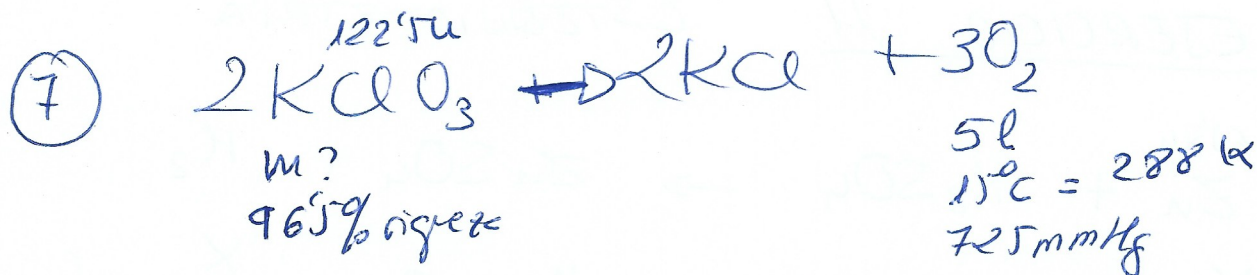
$$\% \text{ Zn} = \frac{0'106}{0'156} \cdot 100 = 67'95\%$$

$$\% \text{ Al} = \frac{0'05}{0'156} \cdot 100 = 32'05\%$$



$$n_{\text{H}_2\text{SO}_4} = n_{\text{H}_2} = \frac{0'156 - X}{18} = \frac{0'05}{18} = 2'78 \cdot 10^{-3}$$

$$2'78 \cdot 10^{-3} = \frac{g_{\text{H}_2\text{SO}_4}}{98} \rightarrow g = 0'272 \text{ g de H}_2\text{SO}_4$$

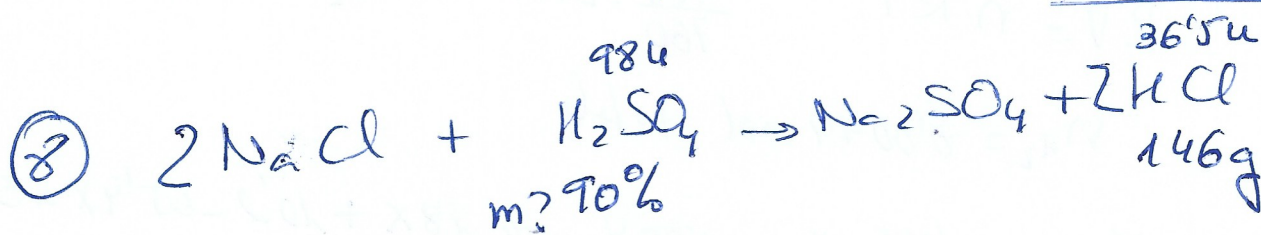


$P \cdot V = n R T$
 $\frac{725}{760} \cdot 5 = n_{\text{O}_2} \cdot 0.082 \cdot 288 \rightarrow n_{\text{O}_2} = 0.2 \text{ mol}$

$\frac{2 \text{ mol KClO}_3}{n_{\text{KClO}_3}} = \frac{3 \text{ mol O}_2}{0.2}$ $n_{\text{KClO}_3} = 0.135 \text{ mol}$

$0.135 = \frac{g_{\text{KClO}_3}}{122.5} \rightarrow g_{\text{KClO}_3} = 16.54 \text{ g pure}$

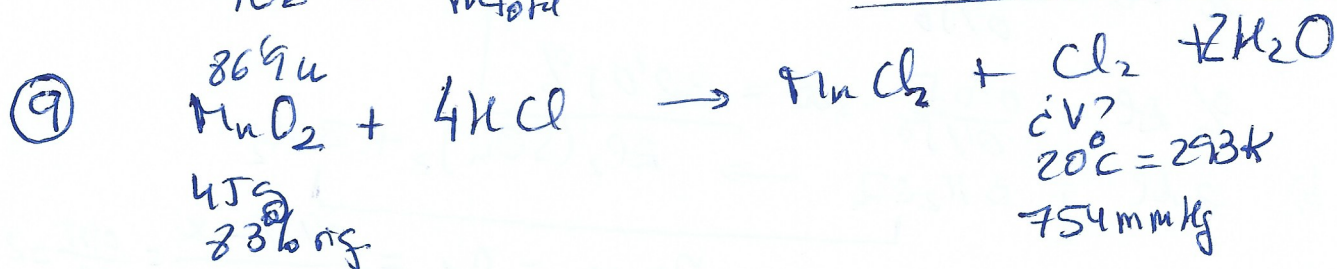
$96.5 = \frac{16.54}{m_{\text{total}}} \cdot 100 \rightarrow m_{\text{total}} = 17.138 \text{ KClO}_3$
 con impurezas



$n_{\text{HCl}} = \frac{146}{36.5} = 4 \text{ mol}$ $n_{\text{H}_2\text{SO}_4} = \frac{n_{\text{HCl}}}{2} = 2 \text{ mol}$

$2 = \frac{g_{\text{H}_2\text{SO}_4}}{98} \rightarrow g_{\text{H}_2\text{SO}_4} = 196 \text{ g H}_2\text{SO}_4 \text{ pure}$

$90\% = \frac{196}{m_{\text{total}}} \cdot 100 \rightarrow m_{\text{total}} = 218.8 \text{ H}_2\text{SO}_4$
 impurezas



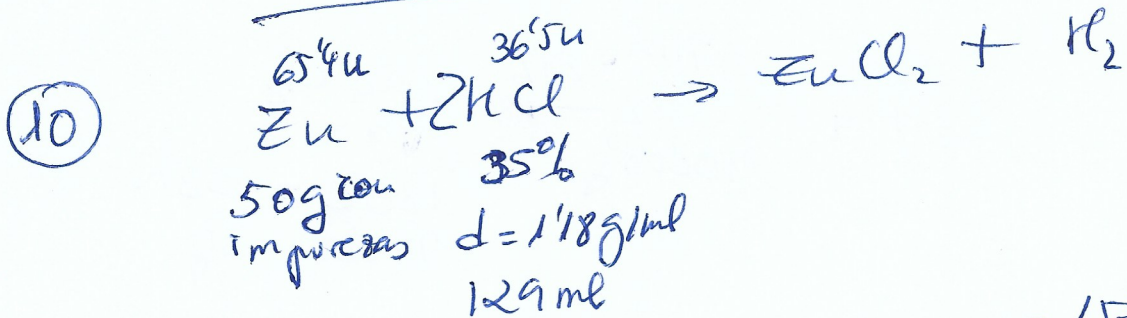
$83 = \frac{m_{\text{MnO}_2}}{45} \cdot 100 \rightarrow m_{\text{MnO}_2} = 37.35 \text{ g pure}$

$n_{\text{MnO}_2} = \frac{37.35}{86.9} = 0.43 \text{ mol}$

$$n_{Cl_2} = n_{HNO_2} = 0.43 \text{ mol}$$

$$P \cdot V = n R T \quad \frac{754}{760} \cdot V_{Cl_2} = 0.43 \cdot 0.0821 \cdot 293$$

$$V_{Cl_2} = 10.41 \text{ l}$$



$$\text{HCl} \rightarrow d = \frac{m_T}{V_T} \quad 1.18 = \frac{m_T}{129} \rightarrow m_T = 152.22 \text{ g}$$

$$35 = \frac{m_{HCl}}{152.22} \cdot 100 \rightarrow m_{HCl} = 53.28 \text{ g}$$

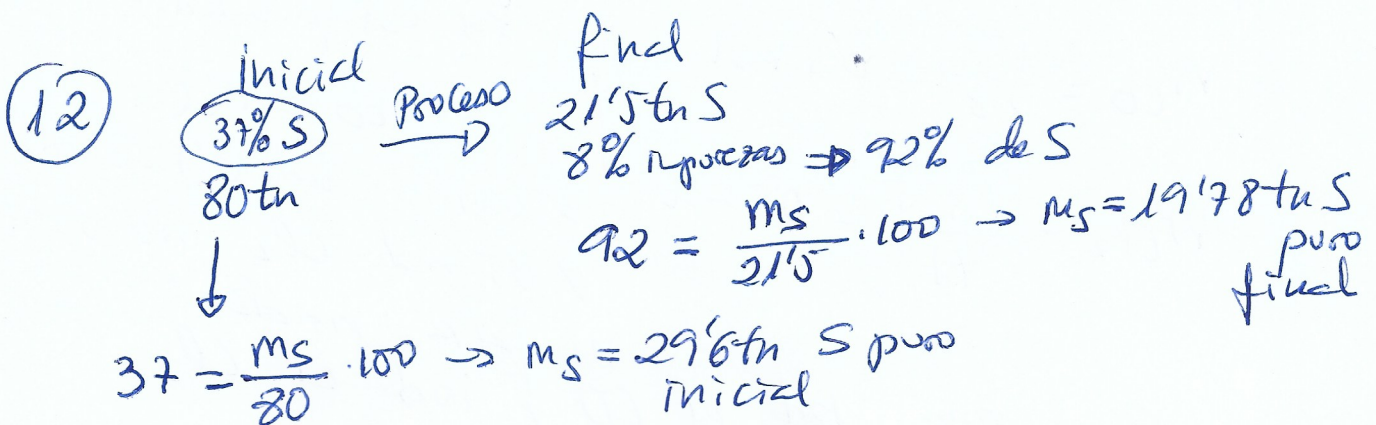
$$n_{HCl} = \frac{53.28}{36.5} = 1.46 \text{ mol}$$

$$\frac{1 \text{ mol Zn}}{n_{Zn}} = \frac{2 \text{ mol HCl}}{1.46} \rightarrow n_{Zn} = 0.73 \text{ mol}$$

$$0.73 = \frac{m_{Zn}}{65.4} \rightarrow m_{Zn} = 47.738 \text{ g Zn puro}$$

$$\% = \frac{47.73}{50} \cdot 100 = 95.46\%$$

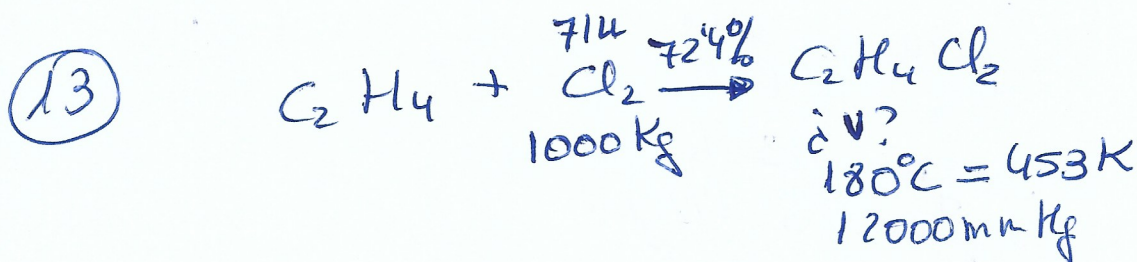
$$M_{HCl} = \frac{n_{HCl}}{V_{total}} = \frac{1.46}{0.129} = 11.32 \text{ mol/l}$$



$$S_{\text{no aprovechado}} = S_{\text{final}} - S_{\text{inicial}} =$$

$$= 29'6 - 19'78 = \underline{9'82 \text{ tu S no aprovechado}}$$

$$\% = \frac{m_{\text{final}}}{m_{\text{inicial}}} \cdot 100 = \frac{19'78}{29'6} \cdot 100 = \underline{66'82\%}$$



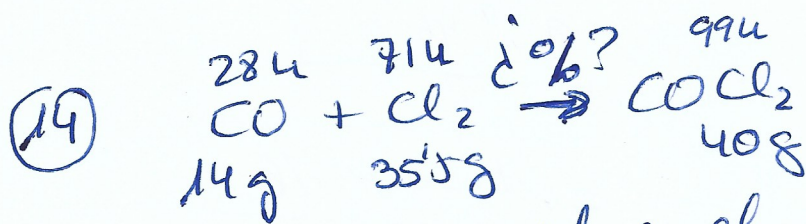
$$n_{\text{Cl}_2} = \frac{1000000}{71} = 14'08 \cdot 10^3 \text{ mol}$$

$$n_{\text{C}_2\text{H}_4\text{Cl}_2} = n_{\text{Cl}_2} = 14'08 \cdot 10^3 \quad P \cdot V = n R T$$

$$\frac{1200}{760} \cdot V = 14'08 \cdot 10^3 \cdot 0'082 \cdot 453$$

$$V = 331'35 \text{ m}^3 \text{ } \Rightarrow \text{reacción total}$$

al 72'4% de rendimiento $331'35 \cdot \frac{72'4}{100} = \underline{240 \text{ m}^3 \text{ de } \text{C}_2\text{H}_4\text{Cl}_2}$



1.º Analizar cuál es el reactivo limitante

$$n_{\text{CO}} = \frac{14}{28} = 0'5 \text{ mol}$$

$$n_{\text{Cl}_2} = \frac{35'5}{71} = 0'5 \text{ mol}$$

Por estequiometría

$$n_{\text{CO}} = n_{\text{Cl}_2}$$

$\rightarrow$ como hay 0'5 mol de CO
 y 0'5 mol Cl_2 no sobra

ni falta nada.

Reacciona todo el CO, todo el Cl_2

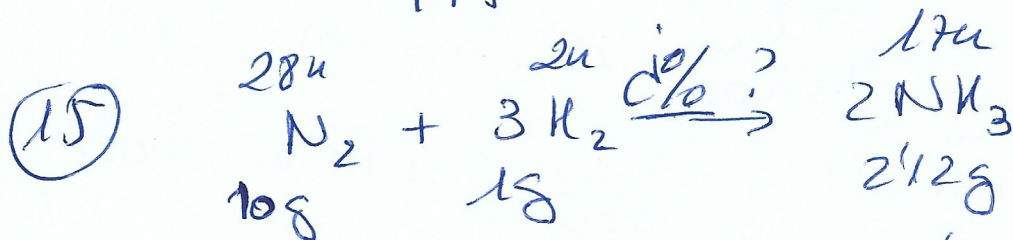
2º calcular el producto

$$n_{\text{COCl}_2} = n_{\text{CO}} = 0.5 \text{ mol}$$

$$0.5 = \frac{m_{\text{COCl}_2}}{99} \Rightarrow m_{\text{COCl}_2} = 49.5 \text{ g si}$$

el rendimiento fuera del 100%

$$\% = \frac{40}{49.5} \cdot 100 = \underline{80.8\%}$$



1º Analizar cual es el reactivo limitante

$$n_{\text{N}_2} = \frac{10}{28} = 0.357 \text{ mol}$$

$$n_{\text{H}_2} = \frac{1}{2} = 0.5 \text{ mol}$$

$$\frac{1 \text{ mol N}_2}{n_{\text{N}_2}} = \frac{3 \text{ mol H}_2}{0.5}$$

$\rightarrow n_{\text{H}_2} = 0.17 \text{ mol de N}_2$
se necesita para gastar
todo el H_2

$$0.17 \text{ mol N}_2 < 0.357 \Rightarrow \text{sobra N}_2$$

Reactivo limitante: H_2

" en exceso: N_2

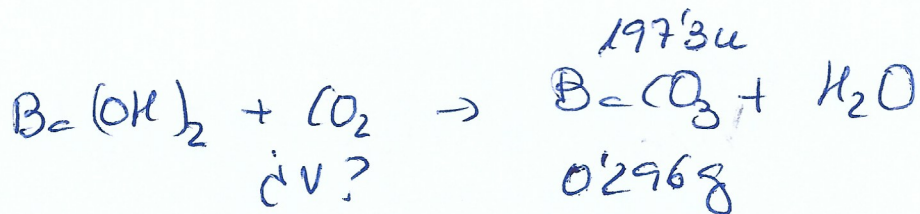
2º Calcular NH_3 con el reactivo limitante

$$\frac{3 \text{ mol H}_2}{0.5} = \frac{2 \text{ mol NH}_3}{n_{\text{NH}_3}} \quad n_{\text{NH}_3} = 0.33 \text{ mol}$$

$$0.33 = \frac{m_{\text{NH}_3}}{17} \Rightarrow m_{\text{NH}_3} = 5.61 \text{ g}$$

$$\% \text{ rendimiento} = \frac{2.12}{5.61} \cdot 100 = \underline{37.4\%}$$

(16)



100 l aire → es una mezcla
20°C = 293K de gases
740 mmHg

$$n_{\text{BaCO}_3} = \frac{0'296}{197'3} = 1'5 \cdot 10^{-3} \text{ mol}$$

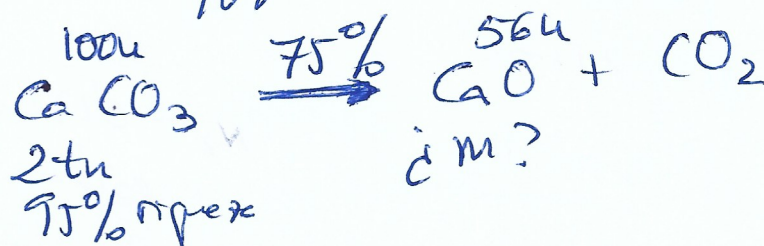
$$n_{\text{CO}_2} = n_{\text{BaCO}_3} = 1'5 \cdot 10^{-3} \text{ mol}$$

$$P \cdot V = nRT \quad \frac{740}{760} \cdot V = 1'5 \cdot 10^{-3} \cdot 0'082 \cdot 293$$

$V = 0'037 \text{ l de CO}_2 \text{ que est\u00e1 en el aire}$

$$\% = \frac{0'037}{100} \cdot 100 = \underline{\underline{0'037\%}}$$

(17)



$$1\text{tn} = 1000 \text{ Kg} = 10^6 \text{ g}$$

$$95 = \frac{m_{\text{CaCO}_3}}{2 \cdot 10^6} \cdot 100 \rightarrow m_{\text{CaCO}_3} = 1'9 \cdot 10^6 \text{ g pure}$$

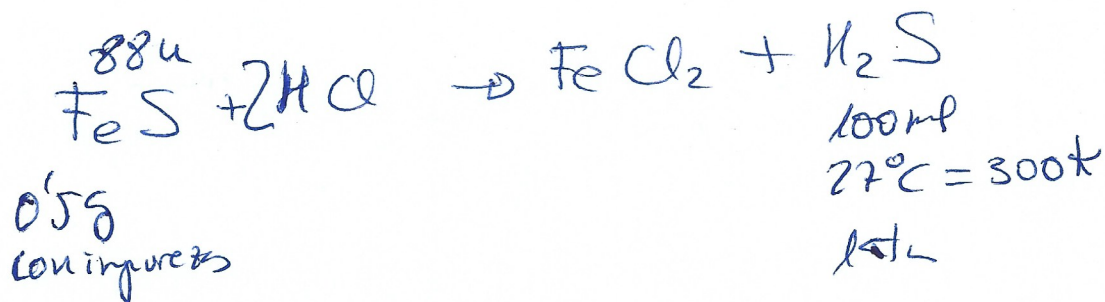
$$n_{\text{CaCO}_3} = \frac{1'9 \cdot 10^6}{100} = 1'9 \cdot 10^4 \text{ mol}$$

$$n_{\text{CaO}} = n_{\text{CaCO}_3} = 1'9 \cdot 10^4 \text{ mol}$$

$$1'9 \cdot 10^4 = \frac{\hat{m}_{\text{CaO}}}{56} \rightarrow \hat{m}_{\text{CaO}} = 1'064 \cdot 10^6 \text{ g si reacciona todo}$$

$$\text{al } 75\% \Rightarrow 1'064 \cdot 10^6 \cdot \frac{75}{100} = 7'98 \cdot 10^5 \text{ g} = \underline{\underline{798 \text{ Kg}}}$$

(18)



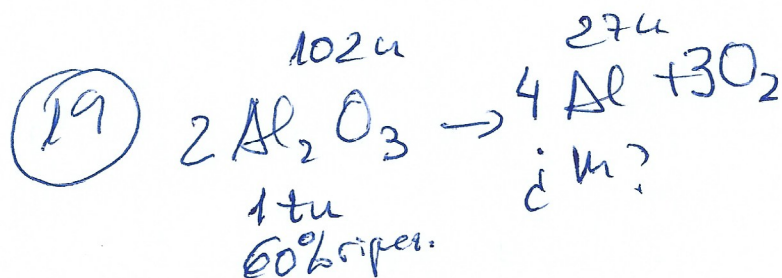
$$P \cdot V = nRT \quad 1. 0.1 = n_{\text{H}_2\text{S}} \cdot 0.082 \cdot 300$$

$$n_{\text{H}_2\text{S}} = 4.06 \cdot 10^{-3} \text{ mol}$$

$$n_{\text{FeS}} = n_{\text{H}_2\text{S}} = 4.06 \cdot 10^{-3} \text{ mol}$$

$$4.06 \cdot 10^{-3} = \frac{g_{\text{FeS}}}{88} \rightarrow g = 0.36 \text{ g pure}$$

$$\% = \frac{0.36}{0.5} \cdot 100 = \underline{\underline{71.5\%}}$$



$$60 = \frac{m_{\text{Al}}}{10^6} \cdot 100 \rightarrow m_{\text{Al}_2\text{O}_3} = 6 \cdot 10^5 \text{ g}$$

$$n_{\text{Al}_2\text{O}_3} = \frac{6 \cdot 10^5}{102} = 5.88 \cdot 10^3 \text{ mol}$$

$$\frac{2 \text{ mol Al}_2\text{O}_3}{5.88 \cdot 10^3} = \frac{4 \text{ mol Al}}{n_{\text{Al}}}$$

$$n_{\text{Al}} = 11.76 \cdot 10^3 \text{ mol}$$

$$11.76 \cdot 10^3 = \frac{g_{\text{Al}}}{27} \rightarrow g_{\text{Al}} = 317.6 \cdot 10^5 \text{ g} = \underline{\underline{3176 \text{ kg}}}$$